

American International University–Bangladesh (AIUB)

Department of Computer Science, Faculty of Science and Technology

CN Lecture 02: Application Layer Protocols

OBE Based Questions and Answers

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1. A university is developing an online learning platform where students access lecture materials through a web browser. Explain how HTTP enables communication between the student's browser and the web server.

Answer:

HTTP (Hypertext Transfer Protocol) is an application-layer protocol used for communication between web browsers (clients) and web servers.

When a student enters a URL, the browser sends an HTTP request to the web server. The server processes the request and returns an HTTP response containing the requested web page or resource.

The communication follows the client-server model.

Steps involved:

- The user enters a URL.
- The browser sends an HTTP request.
- The web server receives and processes the request.
- The server sends an HTTP response.
- The browser displays the received web page.

Thus, HTTP provides a standardized method for exchanging web resources over the Internet.

2. A web application requires users to log in and submit personal information. Compare the GET and POST request methods and recommend the appropriate method with justification.

Answer:

Feature	GET	POST
Purpose	Retrieve data	Submit data
Data Location	URL	Request body
Security	Less secure	More secure
Data Size	Limited	Large amount of data
Typical Use	Viewing web pages	Login forms, registration, file upload

POST should be used because sensitive user information should not be exposed in the URL and larger amounts of data can be transmitted securely.

3. An organization wants to update an employee record through a REST API. Explain the purpose of the PUT request method and distinguish it from GET and POST.

Answer:

PUT is used to create or replace an existing resource on the server.

Comparison:

- GET retrieves existing resources.
- POST submits new data to the server.
- PUT updates or replaces existing resources.

Therefore, PUT is appropriate when an existing employee record must be modified.

4. **A browser cannot open `www.aiub.edu` because it only understands IP addresses. Explain how the Domain Name System (DNS) solves this problem.**

Answer:

DNS (Domain Name System) translates human-readable domain names into IP addresses.

The DNS resolution process is:

- (a) User enters a domain name.
- (b) The browser contacts the Local DNS Server.
- (c) If the address is unavailable locally, the Local DNS contacts the Root DNS Server.
- (d) The Root DNS refers the query to the appropriate Top-Level Domain (TLD) server.
- (e) The TLD server refers the query to the Authoritative DNS Server.
- (f) The Authoritative DNS Server returns the correct IP address.
- (g) The browser uses the IP address to connect to the web server.

DNS allows users to access websites using meaningful names instead of numerical IP addresses.

5. **Compare the roles of Root Name Server, Local Name Server, and Authoritative Name Server in the DNS hierarchy.**

Answer:

DNS Server	Primary Responsibility
Root Name Server	Directs queries to the correct Top-Level Domain server.
Local Name Server	Receives DNS requests from users and performs recursive lookup or caching.
Authoritative Name Server	Stores official DNS records and returns the final IP address.

These servers work together to resolve domain names efficiently.

6. **A network administrator wants to reduce DNS lookup time for frequently visited websites. Explain how the Local DNS Server improves network performance.**

Answer:

The Local DNS Server stores recently resolved domain names in its cache.

Advantages include:

- Faster domain resolution.
- Reduced Internet traffic.
- Lower response time.
- Reduced workload on Root and Authoritative DNS servers.
- Improved browsing performance.

Caching significantly improves the efficiency of DNS resolution.

7. **Differentiate between Recursive DNS Query and Iterative DNS Query with suitable examples.**

Answer:

Feature	Recursive Query	Iterative Query
Responsibility	DNS server completes lookup	Client performs multiple lookups
Client Work	Minimal	Higher
Response	Final answer	Referral to another server
Efficiency	Easier for clients	Reduces DNS server workload

Recursive queries return the final IP address directly, whereas iterative queries return referrals until the requested record is found.

8. Evaluate why HTTP uses TCP instead of UDP for web communication.

Answer:

HTTP uses TCP because reliable communication is essential for web applications.

TCP provides:

- Reliable data delivery.
- Error detection and recovery.
- Packet sequencing.
- Flow control.
- Congestion control.

UDP does not provide these services. Therefore, HTTP relies on TCP to ensure complete and accurate delivery of web pages.

9. A company plans to build an e-commerce website. Analyze the importance of HTTP and DNS working together during website access.

Answer:

DNS and HTTP perform different but complementary functions.

DNS converts the website name into an IP address, allowing the client to locate the server.

HTTP is then used to request and transfer web pages between the browser and the web server.

Without DNS, users would need to remember IP addresses.

Without HTTP, web browsers could not exchange web content with servers.

Therefore, both protocols are essential for accessing modern websites.

10. Analyze the advantages of using domain names instead of IP addresses for Internet communication.

Answer:

Using domain names provides several advantages:

- Easy to remember.
- User-friendly.
- Independent of changing IP addresses.
- Simplifies website management.
- Improves usability for Internet users.

DNS maps these memorable names to numerical IP addresses automatically, making Internet communication easier and more efficient.